Unit 1.3: Measurement Tools Workshop

From measuring everything to mastering what matters in engineering

Why Measurement Matters

Engineers rely on measurement for:

- Quality control
- Fit & function
- Performance verification
- Safety compliance

“If you can’t measure it, you can’t improve it.” — Lord Kelvin

What Could We Measure?

What Could We Measure?

Categories:

- **Physical Dimensions**: length, width, height, diameter, thickness, circumference
- **Shape Features**: radius, chamfer angle, curvature, profile deviation
- **Mass & Weight**: total mass, distribution, center of gravity
- **Position & Distance**: spacing, clearances, offsets, depth

What Could We Measure?

Categories:

- **Angles**: slope, bevel, rotation, tilt
- **Volume**: displacement, capacity, void volume
- **Surface Qualities**: roughness, texture, flatness, waviness
- **Material Properties**: density, hardness, elasticity, tensile strength
- **Environmental**: temperature, humidity, pressure, vibration, noise level

What Could We Measure?

Categories:

- **Electrical**: voltage, current, resistance, capacitance, inductance
- **Time-based**: speed, acceleration, frequency, cycle time
- **Flow-related**: fluid flow rate, air velocity, fuel consumption
- **Optical**: light intensity, color wavelength, reflectivity
- **Human Factors**: grip force, reach distance, reaction time

What Engineers Focus On Most

Dimensions & Geometry (length, width, height, depth, diameter, angles)

Mass & Weight

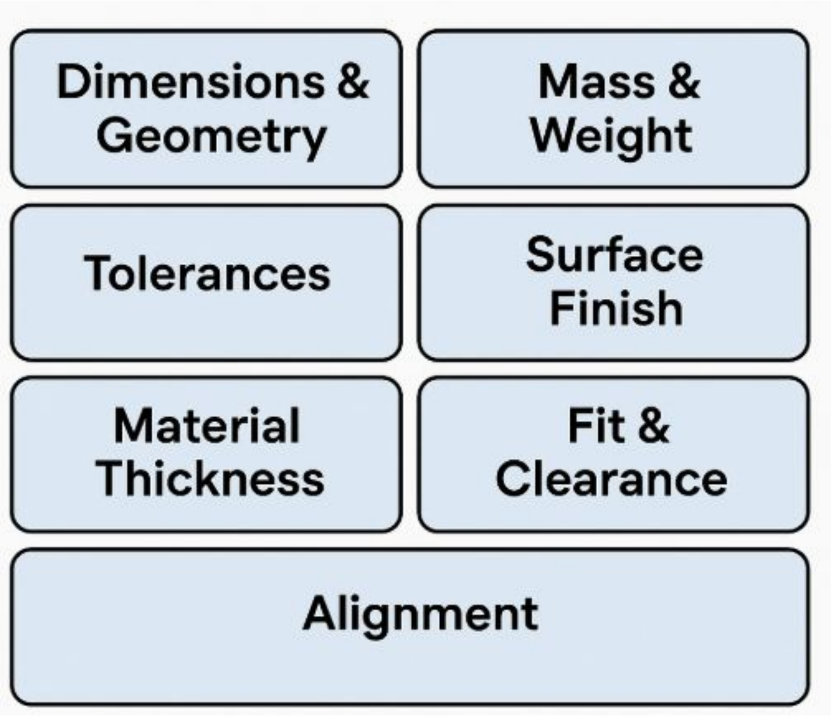
Tolerances (how close a measurement must be)

Surface Finish (roughness, flatness)

Material Thickness

Fit & Clearance (holes, shafts, slots)

Alignment (squareness, concentricity)



The Tools Engineers Use

Dimensional Tools:

- Calipers (digital, dial, vernier)
- Micrometers (outside, inside, depth)
- Rulers & Measuring Tapes
- Height gauges

Mass/Weight Tools:

- Scales & balances

The Tools Engineers Use

Angle Tools:

- Protractors, bevel gauges, inclinometers

Surface Finish Tools:

- Surface roughness testers

Specialty Tools:

- Bore gauges, go/no-go gauges, coordinate measuring machines (CMM)

Tools We Have in Our Lab

Calipers – digital calipers for inside, outside, and depth measurements

Rulers – quick and easy length checks

Measuring Tapes – for larger objects or distances

Scales (Mass) – for weighing parts and assemblies

Hands-On Practice Goals

Select the right tool for the measurement task

Use each tool correctly and safely

Record measurements with correct units and precision

Compare measurements between tools for accuracy

Today’s Activities

Tool Demonstration – Instructor-led

Practice Stations – Rotate through measuring tasks

Measurement Challenge – Guess, measure, and compare results

Exit Ticket – “Tool I’ll use most and why”